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Starter - Answers

1) Simplify:
   a) $3x^2 \times 5x^5$ $15x^7$
   b) $\frac{5x^3y^2}{15x^2y^3}\frac{x}{3y}$

2) Factorise:
   a) $x^2 - 2x - 24$ $(x-6)(x+4)$
   b) $3x^2 - 17x + 20$ $(3x-5)(x-4)$

3) Use long division to calculate:
   a) $197,041 \div 23$ 8567
   b) $56,168 \div 34$ 1652

4) Find the equations of the lines that pass through these pairs of points:
   a) $(-1,4)$ and $(5,-14)$ $y = 1 - 3x$
   b) $(2,-6)$ and $(8,-3)$ $y = \frac{1}{2}x - 7$

5) Complete the square for the expressions:
   a) $x^2 - 2x - 20$ $(x-1)^2 - 21$
   b) $2x^2 + 4x + 15$ $2(x+1)^2 + 13$

1: Algebraic Fractions

2: Dividing Polynomials

3: The Factor Theorem

4: Proof

Recall that you can simplify fractions by **dividing** the numerator and denominator by a **common factor**.

Hint: To identify common factors we need to factorise first.

$$\frac{x^2-1}{x^2+x}$$

$$\frac{x^2+3x+2}{x+1} =$$

Note: Do not leave 1 in the denominator!

Tip: Factorise the easier one first because it provides clues to the factorisation of the other.

$$\frac{2x^2+11x+12}{x^2+9x+20} =$$

$$\frac{4-x^2}{x^2+2x-8} =$$

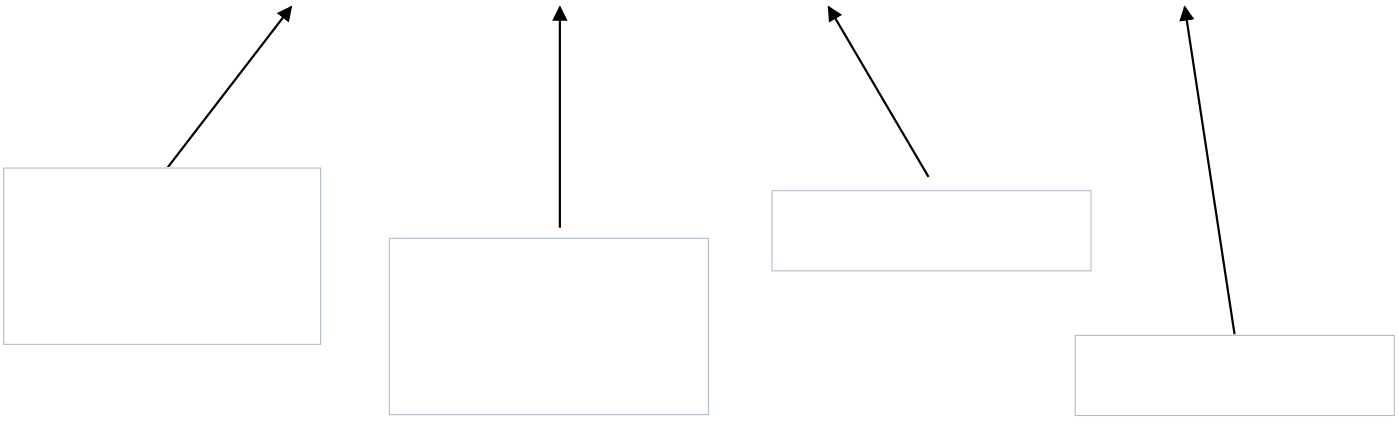
Tip: $\frac{a-b}{b-a} = -1$

2 Simplify these fractions as far as possible:

m $\dfrac{2x^2+3x+1}{x^2-x-2}$ **n** $\dfrac{x^2+6x+8}{3x^2+7x+2}$ **o** $\dfrac{2x^2-5x-3}{2x^2-9x+9}$

3 $\dfrac{6x^3+3x^2-84x}{6x^2-33x+42} = \dfrac{ax(x+b)}{x+c}$, where a, b and c are constants.
Work out the values of a, b and c.

$$11 \div 4 = 2rem3$$



$$
\begin{array}{r}
38. \\
11 \overline{)\,423.0000} \\
33 \\
\hline
93 \\
88 \\
\hline
50
\end{array}
$$

1. We found how many whole number of times (i.e. the quotient) the divisor went into the dividend.

2. We multiplied the quotient by the dividend.

3. . . .in order to find the remainder.

4. Find we 'brought down' the next number.

How many times does x go into $6x^3$?

Now repeat! How many times does x go into $-2x^2$?

$$
\begin{array}{r}
6x^2 - 2x + 3 \\
x + 5 \overline{)\,6x^3 + 28x^2 - 7x + 15} \\
6x^3 + 30x^2 \\
\hline
-2x^2 - 7x \\
-2x^2 - 10x \\
\hline
3x + 15 \\
3x + 15 \\
\hline
0
\end{array}
$$

Multiply $6x^2$ by $(x+5)$. The first term should match with above.

Subtract and carry down next term.

This is the remainder.

Tip:
You can check your solution by expanding:
$(x+5)\left(6x^2-2x+3\right)$
But if you know you should get no remainder, ending with 0 at the bottom is a good sign!

Tip:
Be **very** careful subtracting negatives.
$-7x-(-10x)=3x$

Find the remainder when $3x^3 - 2x + 4$ is divided by $x - 1$.

$$
\begin{array}{r|l}
x - 1 & 3x^2 + 3x + 1 \\
\hline
 & 3x^3 + 0x^2 - 2x + 4 \\
 & 3x^3 - 3x^2 \\
\hline
 & 3x^2 - 2x \\
 & 3x^2 - 3x \\
\hline
 & x + 4 \\
 & x - 1 \\
\hline
 & 5
\end{array}
$$

Note: Ensure that any missing terms in the polynomial are filled in, so that the powers decrease by 1 each time.

The remainder is 5.

Find the remainder when $2x^3 - 5x^2 - 16x + 10$ is divided by $x - 4$.

Divide $8x^3 - 1$ by $2x - 1$.

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15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.
Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.
a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.
b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by
i $x - 2$,
ii $x + 3$.
b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.
a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where a, b and c are constants to be found.
b Hence factorise $\mathrm{f}(x)$ completely.
c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

19 $f(x) = 12x^3 + 5x^2 + 2x - 1$.
a Show that $(4x - 1)$ is a factor of $f(x)$ and write $f(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.
b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

$$x^3 + x^2 - 4x - 4 = (x - 2)\left(x^2 + 3x + 2\right)$$

We can see that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.
What would happen if x is 2?

The Factor Theorem states that if $f(x)$ is a polynomial then:
- If $f(p) = 0$, then $(x - p)$ is a factor of $f(x)$.
- Conversely, if $(x - p)$ is a factor of $f(x)$, then $f(p) = 0$.

Show that $(x-2)$ is a factor of x^3+x^2-4x-4.

Fully factorise $2x^3+x^2-18x-9$.

Given that $2x+1$ is a factor of $6x^3+ax^2+1$, determine the value of a.

$f(x) = 6x^3 + 13x^2 - 4$

(a) Use the remainder theorem to find the remainder when $f(x)$ is divided by $(2x + 3)$. (2)

(b) Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$. (2)

(c) Factorise $f(x)$ completely. (4)

No long in spec.

Given that $3x - 1$ is a factor of $3x^3 + 11x^2 + ax + 1$, determine the value of a.

13 $f(x) = 3x^3 - 12x^2 + 6x - 24$.

a Use the factor theorem to show that $(x - 4)$ is a factor of $f(x)$.

b Hence, show that 4 is the only real root of the equation $f(x) = 0$.

14 $f(x) = 4x^3 + 4x^2 - 11x - 6$.
a Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$.
b Factorise $f(x)$ completely.

15 a Show that $(x - 2)$ is a factor of $9x^4 - 18x^3 - x^2 + 2x$.
b Hence, find four real solutions to the equation $9x^4 - 18x^3 - x^2 + 2x = 0$.

1 *[MAT 2006 1E]* The cubic

$$x^3 + ax + b$$

Has both $(x-1)$ and $(x-2)$ as factors. Determine the values of a and b.

a= −7,b=6

[MAT 2009 1I]

$$n^2x^{2n+6} - 25nx^{n+4} + 150x^7$$

2 Has $x^2 - 1$ as a factor

A) for no values of n;
B) for $n = 10$ only;
C) for $n = 15$ only;
D) for $n = 10$ and $n = 15$ only.

x^2-1= (x+1)(x−1) so both factors.
f (1)=n^2−25n+150=0
If n is odd: f (−1)= −n^2−25n−150=0
If n is even: f (−1)= −n^2+25n−150=0

The **remainder theorem** states that if $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$. This similarly works whenever a makes the divisor 0.

3 [MAT 2013 1G] Let $n \geq 2$ be an integer and $p_n(x)$ be the polynomial

$$p_n(x) = (x-1) + (x-2) + \ldots + (x-n)$$

What is the remainder, in terms of n, when $p_n(x)$ is divided by $p_{n-1}(x)$?

Terminology

A **conjecture** is a mathematical statement that has yet to be proven.

One famous conjecture is **Goldbach's Conjecture**.
It states "*Every even integer greater than 2 can be expressed as the sum of two primes.*"
It has been verified up to 4×10^{18} (that's big!); this provides evidence that it is true, but does not prove it is true!

A **theorem** is a mathematical statement that has been proven.

One famous misnomer was **Fermat's Last Theorem**, which states "*If n is an integer where $n > 2$, then $a^n + b^n = c^n$ has no non-zero integer solutions for a, b, c.*" It was 300 years until this was proven in 1995. Only then was the 'Theorem' in the name then correct!

A proof must show all **assumptions** you are using, have a clear **sequential list of steps** that logically follow, and must cover **all possible cases**. You should usually make a **concluding statement**, e.g. restating the original conjecture that you have proven.

a. Proof by Deduction

This is the simplest type, where you start from known facts and reach the desired conclusion via deductive steps.

An **identity** is an equation that is true for **all values** of the variables. e.g. $x^2 = 4$ is true only for $x = \pm 2$, but $x(x-1) \equiv x^2 - x$ is true for all x.

Proof by Deduction requires you to **start from known facts** and end up at the conclusion. It is **not** acceptable to start with to the conclusion, and verify it works, **because you are assuming the thing you are trying to prove**.

Example: Prove that if three consecutive integers are the sides of a right-angled triangle, they must be 3, 4 and 5.

We are assuming the thing we are trying to prove!

We are only assuming things in the 'if' bit. This is fine!

The underlying problem is that this technique doesn't prove there can't be **other** consecutive integers that work – we have only verified 3,4,5 is one such solution.

a. Proof by Deduction

Exam Tip: This is quite a common last parter.

Prove that x^2+4x+5 is positive for all values of x.

Anything squared is at least 0. This is formally known as the '*trivial inequality*'.

Test Your Understanding

Prove that the sum of the squares of two consecutive odd numbers is 2 more than a multiple of 8.

$2n-1$ $2n+1$

11 Prove that $x^2 + 8x + 20 \geq 4$ for all values of x.

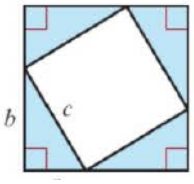
12 The equation $kx^2 + 5kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < \frac{12}{25}$.

13 The equation $px^2 - 5x - 6 = 0$, where p is a constant, has two distinct real roots. Prove that p satisfies the inequality $p > -\frac{25}{24}$.

18 A circle has equation $(x-1)^2 + y^2 = k$, where $k > 0$.
The straight line L with equation $y = ax$ cuts the circle at two distinct points.
Prove that $k > \dfrac{a^2}{1+a^2}$.

19 Prove that the line $4y - 3x + 26 = 0$ is a tangent to the circle $(x+4)^2 + (y-3)^2 = 100$.

20 The diagram shows a square and four congruent right-angled triangles.
Use the diagram to prove that $a^2 + b^2 = c^2$.



[STEP I 2005 Q1] 47231 is a five-digit number whose digits sum to $4 + 7 + 2 + 3 + 1 = 17$.

(i) Prove that there are 15 five-digit numbers whose digits sum to 43. You should explain your reasoning clearly.

(ii) How many five-digit numbers are there whose digits sum to 39?

b. Proof by Exhaustion

This means breaking down the statement into **all possible smaller cases**, where we prove each individual case.

(This technique is sometimes known as 'case analysis')

c. Disproof by Counter-Example

While to prove a statement is true, we need to prove every possible case (potentially infinitely many!), **we only need one example to disprove** a statement.

This is known as a **counterexample**.

Disprove the statement:
"$n^2 - n + 41$ is prime for all integers n."

4 Prove that all cube numbers are either a multiple of 9 or 1 more or 1 less than a multiple of 9.

7 Prove that for all real values of x
$(x + 6)^2 \geq 2x + 11$

8 Given that a is a positive real number, prove tha
$a +$

9 a Prove that for any positive numbers p and q:
$p + q > \sqrt{4pq}$
b Show, by means of a counter-example, that this inequality does not hold when p and q are both negative. (2 m . . .